



# Anas TBER

## Data Science & AI Engineering Student — Centrale Lyon

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Engineering student at Centrale Lyon with hands-on experience in graph-based ML, NLP, and large-scale data engineering. Seeking an **end-of-studies internship in Data Science or AI** starting **March 2027**.

### Education

#### École Centrale de Lyon — General Engineering

Lyon · Sep. 2023 – Apr. 2027

- **Courses:** Machine Learning & Applications, Collaborative Algorithms, Probability & Stochastic Processes, Algorithms for Reasoning, OOP, Enterprise Management

#### Lycée Henri Loritz — CPGE, PCSI / PSI\*

Nancy · Sep. 2020 – Jun. 2023

- Intensive preparatory classes in Physics, Mathematics & Engineering Sciences for France's top engineering school entrance exams (Grandes Écoles).

### Experience

#### Data Scientist Intern — BNP Paribas CIB — Compliance, Anti-Money Laundering

Paris · Jan. 2026 – Jul. 2026

- Delivered a **regulator-approved** Networj based AML fraud-detection & investigation system covering the full cycle from raw data ingestion to compliance dashboards, operating on a **30M+ entity** transaction dataset
- Modeled transaction networks using graph algorithms (BFS, DFS, Union-Find) for anomaly detection; designed a **cycle detection** system to identify money mule networks and circular transaction flows
- Engineered graph-based features (degree centrality, cycle membership, neighbor risk scores) to surface high-risk entities across multiple watchlists (Sayari, WikiRan, Crypto, bank sensitivity)
- Processed and validated **100M+ financial records** ensuring data integrity and regulatory compliance using **Polars** and **PyArrow**; built an interactive **Streamlit** dashboard for compliance analysts

#### Tech & Operations Intern — Gentle Mates — Professional Esports (Squeezie, Gotaga)

Paris · Apr. 2025 – Nov. 2025

- Built an automated social media analytics dashboard (Streamlit/Python) integrating computer vision (**YOLO**) for sponsor logo detection and NLP (**Regex**) for brand mention tracking, improving partnership ROI visibility
- Contributed to the development of impactful software and AI internal tools for analytics, monitoring, and operations

### Academic & Personal Projects

#### Survival Analysis — Acute Myeloid Leukemia — Challenge Data ENS by QRT

2025

- **LightGBM** survival model achieving **0.7745 IPCW-C-index** for patient risk stratification on multi-center clinical data; applied feature engineering on genomic datasets with optimized hyperparameters for censored data

#### Transformer Architectures & NLP — Research Project

2024

- Sentiment analysis with **CamemBERT** — **95.91% accuracy** on French reviews; studied attention mechanisms, Word2Vec/GloVe/BERT embeddings, and Transformer vs. **Mamba** (SSM) complexity trade-offs

#### Ant Colony Optimization — Shortest Path Solver

2024

- Implemented **ACO** metaheuristic in Python (OOP) with pheromone trails and probabilistic decision-making; built a real-time **Tkinter** visualization of pathfinding convergence

### Skills

|                   |                                                                        |
|-------------------|------------------------------------------------------------------------|
| Programming       | Python, JavaScript                                                     |
| AI & Data Science | Pandas, NumPy, Scikit-Learn, PyTorch, LightGBM, LangChain, Jupyter     |
| Data Engineering  | Polars, PyArrow, NetworkX, SQL, PostgreSQL                             |
| Web & Viz         | Streamlit, React, HTML/CSS                                             |
| DevOps            | Git, GitHub, GitLab                                                    |
| Languages         | French (Native), Arabic (Native), English (Fluent), Spanish (Beginner) |